

Intro to AedesTraits

Huxley et al.

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Goal:

- To synthesise all of the available data on the temperature dependence of *Aedes* traits into a singel dataset that is open and fully accessible to the VBD community.

Data: Temperature–trait data from the AedesTraits dataset

- **Context:** Invasive *Aedes* mosquitoes (e.g. *Ae. aegypti*, *Ae. albopictus*) drive arboviral disease spread; temperature strongly affects vector traits.
- **Problem:** Trait data are abundant but scattered, inconsistent, and hard to synthesise — especially for emerging species (*Ae. japonicus*, *Ae. koreicus*).
- **Solution:** Developed AedesTraits, an open-access, machine-readable database aligned with VecTraits standards.
- **Content:** 135 digitised studies, ~31,800 records of temperature–dependent traits across life-history, morphology, physiology, behaviour, and transmission.
- **Impact:** Identifies knowledge gaps, supports mechanistic vector modelling, harmonises data, and enables forecasting tools for arbovirus outbreaks.

Data: Temperature–trait data from the AedesTraits dataset

<https://zenodo.org/records/15149904>





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Dataset

 Open

AedesTraits

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AedesTraits is a fully open, machine-readable database that adheres to VecTraits standards, focusing on the temperature dependence of different types of traits (e.g., life–history, behavioural, morphological) in four *Aedes* species: *Ae. aegypti*, *Ae. albopictus*, *Ae. japonicus*, and *Ae. koreicus*. This database consolidates experimental data from diverse published literature dating from 1930s to 2025. **AedesTraits** captures trait variation across species, populations, and environmental conditions.

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AedesTraits: A global database of temperature-dependent trait responses in *Aedes* mosquitoes

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AedesTraits: A global database of temperature-dependent trait responses in *Aedes* mosquitoes

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AedesTraits: A global database of temperature-dependent trait responses in *Aedes* mosquitoes

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Abstract